

Wiring Guide

Every connection, given as the actual perfboard hole (column letter + row number, matching the printed grid on your board) — not an abstract header-pin number. Read a grid ref, find that hole, solder.

Power (+5V / +3V3 / VIN)

Ground

I²C bus (SCL / SDA / IRQ)

Signal (per-pin GPIO)

Teensy header pinout

Two 24-pin female sockets, row A on hole-row **25** , row B on hole-row **31** . Every cell shows the real board hole under that header pin.

Header A — row 25														Teensy pins 25–48	
A25 +5V VIN	Z25 GND —	Y25 +3V3 (out)	X25 D23 B4.2	W25 D22 B3.2	V25 D21 B2.2	U25 D20 B1.2	T25 SCL bus	S25 SDA bus	R25 IRQ3 S3.1	Q25 IRQ2 S2.1	P25 IRQ1 S1.1	O25 D14 CB5.1	N25 D13 CB4.1	M25 GND —	

Header B — row 31															Teensy pins 1–24	
A31 GND —	Z31 D0 J1.1	Y31 D1 J1.2	X31 D2 J1.3	W31 D3 J1.4	V31 D4 J2.1	U31 D5 J2.2	T31 D6 J2.3	S31 D7 J2.4	R31 D8 R1.2	Q31 D9 R2.2	P31 D10 CB1.1	O31 D11 CB2.1	N31 D12 CB3.1	M31 +3V3 —		

Shared bus rails

Run each as one continuous wire past every hole listed, rather than home-running each connector back to the header separately.

GND

Header A: Z25 / M25 · Header B: A31 · feeds every B/CB/J/S connector's ground pin

+3V3

Header A: Y25 · Header B: M31 · feeds S1–S6 pin 5, R12.2, R13.2

+5V

Header A: A25 (VIN) · feeds S1–S6 pin 4

SCL

Header A: T25 · feeds S1–S6 pin 2, R13.1

SDA

Header A: S25 · feeds S1–S6 pin 3, R12.1

S1–S6 — haptic drivers (I²C)

Rotated vertical, one per column. GND/SCL/SDA/+5V/+3V3 all fall on the same row across all six — a horizontal bus wire along that row reaches every connector.

S1		JST-XH-6
1	D11	IRQ1 → P25
2	D10	SCL → T25 (rail)
3	D9	SDA → S25 (rail)
4	D8	+5V → A25 (rail)
5	D7	+3V3 → Y25 (rail)
6	D6	GND → Z25/M25 (rail)

S2		JST-XH-6
1	Y11	IRQ2 → Q25
2	Y10	SCL → T25 (rail)
3	Y9	SDA → S25 (rail)
4	Y8	+5V → A25 (rail)
5	Y7	+3V3 → Y25 (rail)
6	Y6	GND → Z25/M25 (rail)

S3		JST-XH-6
1	T11	IRQ3 → R25
2	T10	SCL → T25 (rail)
3	T9	SDA → S25 (rail)
4	T8	+5V → A25 (rail)
5	T7	+3V3 → Y25 (rail)
6	T6	GND → Z25/M25 (rail)

S4		JST-XH-6
1	011	IRQ4 → J25
2	010	SCL → T25 (rail)
3	09	SDA → S25 (rail)
4	08	+5V → A25 (rail)
5	07	+3V3 → Y25 (rail)
6	06	GND → Z25/M25 (rail)

S5		JST-XH-6
1	J11	IRQ5 → K25
2	J10	SCL → T25 (rail)
3	J9	SDA → S25 (rail)
4	J8	+5V → A25 (rail)
5	J7	+3V3 → Y25 (rail)
6	J6	GND → Z25/M25 (rail)

S6		JST-XH-6
1	E11	IRQ6 → L25
2	E10	SCL → T25 (rail)
3	E9	SDA → S25 (rail)
4	E8	+5V → A25 (rail)
5	E7	+3V3 → Y25 (rail)
6	E6	GND → Z25/M25 (rail)

B1–B4 & CB4/CB5 & R3–R5

B1		JST-XH-2
1	A16	GND (rail)
2	A15	D20 → U25

B2		JST-XH-2
1	V16	GND (rail)
2	V15	D21 → V25

B3		JST-XH-2
1	R16	GND (rail)
2	R15	D22 → W25

B4		JST-XH-2
1	M16	GND (rail)
2	M15	D23 → X25

CB4		JST-XH-4
1	B23	D13 → N25
2	B22	GND (rail)
3	B21	LED_CONF → R4.1
4	B20	GND (rail)

CB5		JST-XH-4
1	V23	D14 → O25
2	V22	GND (rail)
3	V21	LED_ALT → R5.1
4	V20	GND (rail)

R3		220Ω · LED_SFT
1	P20	← CB3.3
2	L20	D36 → G25

R4		220Ω · LED_CONF
1	K20	← CB4.3
2	G20	D37 → H25

R5		220Ω · LED_ALT
1	E20	← CB5.3
2	A20	D38 → I25

J1/ J2 — joysticks

J1		JST-XH-5 · left stick
1	Y39	D0 → Z31
2	Y38	D1 → Y31
3	Y37	D2 → X31
4	Y36	D3 → W31
5	Y35	GND (rail)

J2		JST-XH-5 · right stick
1	039	D4 → V31
2	038	D5 → U31
3	037	D6 → T31
4	036	D7 → S31
5	035	GND (rail)

CB1–CB3 & R1/R2 & B5–B8 & R12/R13

CB1		JST–XH–4
1	D46	D10 → P31
2	D45	GND (rail)
3	D44	LED_ARM → R1.1
4	D43	GND (rail)

CB2		JST–XH–4
1	W46	D11 → Q31
2	W45	GND (rail)
3	W44	LED_EN → R2.1
4	W43	GND (rail)

CB3		JST–XH–4
1	P46	D12 → N31
2	P45	GND (rail)
3	P44	LED_SFT → R3.1
4	P43	GND (rail)

R1		220Ω · LED_ARM
1	K39	← CB1.3
2	G39	D8 → R31

R2		220Ω · LED_EN
1	E39	← CB2.3
2	A39	D9 → Q31

B5		JST–XH–2
1	B52	GND (rail)
2	B51	D24 → L31

B6		JST–XH–2
1	W52	GND (rail)
2	W51	D25 → K31

B7		JST–XH–2
1	S52	GND (rail)
2	S51	D26 → J31

B8		JST–XH–2
1	N52	GND (rail)
2	N51	D27 → I31

R12		4.7kΩ · SDA pull-up
1	J50	SDA (rail)
2	F50	+3V3 (rail)

R13		4.7kΩ · SCL pull-up
1	E50	SCL (rail)
2	A50	+3V3 (rail)

NP1 — numpad

NP1		JST–XH–8 · numpad
1	B30	D28 → H31
2	B29	D29 → G31
3	B28	D30 → F31
4	B27	D31 → E31
5	B26	D32 → D31
6	B25	D33 → D25
7	B24	D34 → E25
8	B23	D35 → F25

Derived directly from the schematic netlist and the final `haptic-console-control-unit-perfboard.kicad_pcb` placement. GND and +5V are fully available on the main header (pins 1/34/47 and 48/VIN) — the Teensy's USB-Host and Ethernet auxiliary pads are not needed. Grid refs use the board's own printed columns (F E D C B A Z Y X...G F E D C B A) and rows (01–53).